THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–8: **Adolescence: A Stage of Growth and Change, Heat Transfer in Nature, Measurement of Time and Motion, Life Processes in Animals** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** growth rates vary and the cousin may have started his spurt	Individuals grow differently.
2	**(a)** A is a better conductor	Faster fall means faster conduction.
3	**(c)** 20 m/s	(150+250)/20.
4	**(b)** fat-rich food	Bile aids fat digestion.
5	**(c)** larynx is growing and adjusting	Settles in time.
6	**(a)** reversed	Land breeze.
7	**(a)** 15 km	60 - 45.
8	**(b)** 1,15,200 times	80 x 60 x 24.
9	**(b)** a rapid rise during the growth spurt	Sudden increase.
10	**(a)** sea breeze	Fluid motion.
11	**(b)** 1.82 s	Mean 18.2 / 10.
12	**(a)** to provide a large surface area for absorption	Large area.
13	**(a)** Both A and R are true and R is the correct explanation of A.	Growth spurts and other changes of the teenage years are brought about by hormones.
14	**(a)** Both A and R are true and R is the correct explanation of A.	R describes the result of the flow from hot to cold, supporting A.
15	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the relation between minute and second does not explain why the second is the SI unit.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Hormones carry messages through blood (1); They control growth and mood (1).	1+1
17	Heating from the top gives little convection (1); From the bottom, hot liquid rises and cooler sinks (1).	1+1
18	Distance 500 m in 30 s (1); 500/30 = 16.67 m/s, i.e. 60 km/h (1).	1+1
19	Respiration releases energy as heat (1); Metabolism increases after a meal (1).	1+1
20	Testosterone promotes muscle growth (1); Greater rise in boys (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Growth hormone lacking (1); Poor growth (1); Other glands underactive (1).	1+1+1
22	Thermocol best insulator (1); Ceramic next (1); Metal loses most heat (1).	1+1+1
23	Speed = 120/12 = 10 m/s (1); Distance in 30 s = 300 m (1); Platform = 300 - 120 = 180 m (1).	1+1+1
24	Fat digestion poor (1); Fewer enzymes (1); Special diet needed (1).	1+1+1
	OR: Avoid excess sweets (1); Carry medicine (1); Eat at regular times (1).	1+1+1
25	Steel conducts (1); Plastic does not (1); Steel handle becomes hot (1).	1+1+1
	OR: Time = 3 h (1); Distance = speed x time (1); 40 x 3 = 120 km (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) Puberty (adolescence) (1). (ii) Testosterone (1). (iii) The testes (1). (iv) The voice box (larynx) grows larger at puberty (1), so the voice becomes deeper; in boys the enlarged voice box can be seen as the Adam's apple (1).	1+1+1+2
27	0-10 s: 2 m/s (1); 10-15 s: rest (1); 15-20 s: 6 m/s (1); Average 50/20 = 2.5 m/s (1); Graph has a flat part (1).	1+1+1+1+1

	OR: Rest: count breaths per minute (1); After walking (1); After running (1); Table of results (1); Rate rises with activity (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–8: **Adolescence: A Stage of Growth and Change, Heat Transfer in Nature, Measurement of Time and Motion, Life Processes in Animals** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** digestion of all three food types	Pancreatic juice is important.
2	**(c)** menstruation	Ovary hormones needed.
3	**(d)** nearly stops	Temperatures equalise.
4	**(c)** 360 km	Gap 120 km, closing 30 km/h in 4 h.
5	**(b)** 8000 mL	16 x 500.
6	**(b)** around 11 to 13 years	Growth spurt.
7	**(a)** heating iron rod	Solid transfer.
8	**(b)** less than true by about 0.2 s	Late start.
9	**(b)** plant food takes longer to digest	Cellulose.
10	**(b)** anaemia	Iron lack.
11	**(c)** hot air rising pushes it	Convection currents.
12	**(c)** 4 m/s	400/100.
13	**(a)** Both A and R are true and R is the correct explanation of A.	The enzyme in saliva starts acting on starch in the mouth, so R explains A.
14	**(c)** A is true but R is false.	The pituitary controls other glands, so A is true; insulin is made by the pancreas, so R is false.
15	**(a)** Both A and R are true and R is the correct explanation of A.	Only radiation can travel through empty space, so R explains A.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Evaporation takes heat (1); Keeps water cool (1).	1+1
17	Time spent at each speed differs (1); Average speed = total distance/total time (1).	1+1
18	Right sends blood to lungs nearby (1); Left pumps to the whole body (1).	1+1
19	Hormonal changes affect mood (1); Support prevents stress (1).	1+1
20	Blanket is an insulator (1); Slows heat entering (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Move to site (1); Engulf germs (1); Defence (1).	1+1+1
22	Alertness (1); Energy (1); But excess causes stress (1).	1+1+1
23	Trap air (1); Insulation (1); Body heat is retained (1).	1+1+1
24	1700/340 (1); 5 s (1); Distance/speed (1).	1+1+1
	OR: Adds bulk (1); Speeds movement (1); Prevents constipation (1).	1+1+1
25	Vitamins deficiency (1); Add fruits and vegetables (1); Reduce fried food (1).	1+1+1
	OR: Hot air rises above (1); Small windows reduce radiation (1); Thick walls slow conduction (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) Conduction (1). (ii) Wood is a poor conductor (insulator) of heat (1). (iii) Convection (1). (iv) Water at the bottom gets heated, becomes lighter and rises (1); cooler, heavier water from the top moves down and a convection current heats all the water (1).	1+1+1+2

27	Distance 80 km in 2 h (1); Average 40 km/h (1); Speedometer shows instantaneous (1); Values differ (1); Both useful (1).	1+1+1+1+1
	OR: Lung damage (1); Blood vessel damage (1); Heart disease (1); Toxins strain kidneys (1); Avoid smoking (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–8: **Adolescence: A Stage of Growth and Change, Heat Transfer in Nature, Measurement of Time and Motion, Life Processes in Animals** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 12 km	x/4 + x/6 = 5 gives x = 12.
2	**(b)** 480 litres	8 L x 60.
3	**(a)** it can occur from about 10 to 15 years	Normal range.
4	**(a)** solar cooker	Absorbs sunlight.
5	**(b)** up to 5 per cent	0.1/2.
6	**(c)** shorter	Meat is easier to digest.
7	**(a)** swelling of the neck	Goitre.
8	**(b)** radiation	Direct radiation.
9	**(a)** 200 m	Back at the start.
10	**(d)** poor absorption of digested food	Less surface area.
11	**(a)** rise	No control.
12	**(c)** heats and cools quickly	Low heat capacity.
13	**(a)** Both A and R are true and R is the correct explanation of A.	R restates the effect of length on the time period and explains A.
14	**(c)** A is true but R is false.	Bile helps in fat digestion, so A is true; but bile contains no enzymes (it only breaks fat into tiny droplets), so R is false.
15	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but one fact does not explain the other.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	The speed stays constant (1); Equal distances are covered in equal times (1).	1+1
17	The air in the alveoli has more oxygen than the blood (1); Oxygen moves into the blood (1).	1+1
18	Heredity and nutrition differ (1); Normal variation (1).	1+1
19	Metal conducts heat to the fins (1); Black radiates heat well (1).	1+1
20	Mean total = 15.2 s (1); T = 1.52 s (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Take 120 km each way (1); Times 3 h and 2 h (1); 240/5 = 48 km/h (1).	1+1+1
22	Muscles need more oxygen (1); Heart pumps more (1); Blood removes CO2 faster (1).	1+1+1
23	Sunlight gives vitamin D (1); Helps calcium absorption (1); Strong bones (1).	1+1+1
24	Coolant absorbs heat inside (1); Releases it outside (1); Back coils warm (1).	1+1+1
	OR: Time period 2 s (1); Each swing 1 s (1); Counting swings (1).	1+1+1
25	Insulin is low (1); Blood sugar rises (1); Control diet (1).	1+1+1
	OR: Stand up politely (1); Explain it is natural (1); Inform teacher (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	(i) $60/30 = 2$ s (1). (ii) No, it does not (1). (iii) $50/20 = 2.5$ s (1). (iv) The second pendulum is longer (1), because a longer pendulum has a larger time period (2.5 s is greater than 2 s) (1).	1+1+1+2
27	Lengths 25, 50, 75, 100 cm (1); 20 oscillations timed each (1); T = t/20 (1); T increases with length (1); Mass kept same (1).	1+1+1+1+1

	OR: Heart pumps blood into arteries (1); Arteries carry it to organs (1); Capillaries exchange materials with cells (1); Veins bring blood back (1); Blood returns to the heart (1).	1+1+1+1+1

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